

Course Syllabus

Course Code:	CPE334
Course Name:	Introduction to Software Engineering in the Age of AI Agents
Course Credit:	3 (3-0-6)
Semester/Year:	1/2026
Prerequisite:	-
Class Meeting:	Lecture & Lab Sections 1, 2, HDS: Tuesdays 13:30 – 17:20 Sections 31, 32: Thursdays 08:30 – 12:20
Class Website:	https://leb2.kmutt.ac.th
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Course Description:	This course introduces the core principles and practices of software engineering in a development environment where AI agents and LLMs assist with design specifications, coding, testing, debugging, refactoring, documentation, and deployment. The central message is that AI tools do not replace software engineering discipline: students must still understand requirements, design software, write specifications, create tests, manage code changes, review work, secure applications, deploy systems, and improve their process. The AI coding agent is treated as a fast but accountable junior developer that requires clear instructions, tests, review, and human accountability. The first half builds individual capability through four structured labs, in which each student develops their own Service Desk / Issue Tracker web application as a platform for practicing the full engineering workflow. The second half is team-based: students build one substantial project from an approved list using Agile project management, code reviews, CI/CD, cloud deployment, testing, documentation, and final delivery.
Learning Outcome:	After completing this course, the student should be able to CLO1 — Build individually. Build and demonstrate a working full-stack web application on your own, from an empty Git repository to a tested, running app, using an AI coding agent responsibly. Evaluated by: the individual labs (TokTickIT) — repo, passing tests, and a working demo. CLO2 — Work as a team. Work in an Agile team to plan, build, and deliver a substantial software product using shared Git workflow, code reviews, and sprints. Evaluated by: the group project — proposal, Kanban board, reviewed PRs, and final delivery. CLO3 — Apply engineering practices. Apply core software-engineering practices — clear requirements, documentation (Spec DD and Test DD) and tests, basic security, automated testing, and simple CI/CD with cloud deployment. Evaluated by: exams plus lab and project artifacts (spec/test docs, security, CI/CD, deployment).
Teaching Method:	Lectures, structured hands-on labs, and team-based project-based learning with AI coding agents
Student Evaluation:	Midterm Exam: 20% Final Exam: 20% Labs / Assignments: 35% Group Project: 25%
Reference:	To be announced. Course materials are distributed via LEB2 (https://leb2.kmutt.ac.th) and the class Facebook group “CPE334 Software Engineering in the Age of AI Agents.”

Class Policy: Students are responsible for all announcements and changes made in class. Academic integrity and the honesty policy will be strictly enforced. AI coding agents may be used, but students remain fully accountable for all submitted work and must document their AI usage.

Course Schedule

The following topics will be covered in our schedule. The instructor may revise parts of the outline to conform to the background, knowledge, and interests of the students.

Week	Date	Topics	Activities
1	4,6 Aug	Course Introduction: Software Engineering in the Age of AI Coding Agents (Dr. Piyanit)	Lab: Lab 1 (Week 1 of 2) — Project Foundation & Vertical Slice: GitHub repo, Git Flow, Issues/Kanban, project scaffold (React/Vite/Bootstrap → Express → Prisma/PostgreSQL)
2	11,13 Aug	Requirements Engineering, Proposal Thinking, Traceability, and Feature Scoping (Dr. Piyanit)	Lab: Lab 1 (Week 2 of 2) — API health check, category seed & list, automated tests (Vitest/Supertest), peer-reviewed PR → lab1-staging → main
3	18,20 Aug	Specification, Architecture, Tech Stack, Data Modeling, and Lightweight UML (Dr. Suthep)	Lab: Lab 2 (Week 1 of 2) — Ticket Creation: engineering contract, Spec DD, Test DD, DB increment & seed, REST API contract
4	25,27 Aug	Test Design, TDD, Verification, Automated Testing, and Definition of Done (Dr. Suthep)	Lab: Lab 2 (Week 2 of 2) — Create Ticket / My Tickets / Ticket Detail screens, attachments, Zen Green theme UI, full test suite (unit/API/UI/E2E)
5	1,3 Sep	Software Security: Authentication, Passwords, Sessions, User Management; Authorization, Roles, SQL Injection, and Common Vulnerabilities (Dr. Piyanit)	Lab: Lab 3 (Week 1 of 2) — Authentication, mandatory first-login password change, User model, migration from Development Requester
6	8,10 Sep	Project Management I: Work Breakdown, Estimation, Scheduling, Resource Constraints, and Definition of Ready (Dr. Santawat)	Lab: Lab 3 (Week 2 of 2) — Role-based authorization (Requester / IT Staff / Administrator), IT Staff ticket queue, Administrator user management, public comments & internal notes
7	15,17 Sep	Team Project Kickoff: Proposal, Backlog, Agile Planning, Team Workflow, and Repository Hygiene (Dr. Santawat)	Lab: Lab 4 (Week 1 of 2) — Actions Taken model, Ticket status-transition matrix, cross-record business rules

Week	Date	Topics	Activities
8	22,24 Sep	Architecture, Infrastructure, API Design, UML, and Test Strategy for Team Projects (Dr. Santawat)	Lab: Lab 4 (Week 2 of 2) — IT Staff & Requester dashboards, final regression & product hardening (complete TokTickIT)
9	6,8 Oct	CI/CD, Build Automation, Quality Gates, and Release Discipline (Dr. Santawat)	Lab: Team Project Kickoff + CI/CD — set up the team repo, proposal & backlog, branch protection, and a CI pipeline with build and automated-test quality gates on every Pull Request
10 (Holiday)	13,15 Oct	Midterm Review and Individual Practical Integration (Dr. Suthep)	Lab: Individual Practical Challenge & Demo (midterm integration); begin Team Sprint 1 — working vertical slice
	14–28 Oct	Midterm Exam (covers Weeks 1–7)	
11	3,5 Nov	Cloud Deployment and Delivery to Users (Dr. Suthep)	Lab: Cloud Deployment Lab — deploy the team application to the cloud (GCP) and deliver to users; Team Sprint 2
12	10,12 Nov	Quality Assurance, UAT, Defect Management, Maintenance, and Technical Debt (Dr. Piyanit)	Lab: QA & UAT — write UAT scripts, defect triage & fixing, maintenance tickets, and a technical-debt log; Team Sprint 3
13	17,19 Nov	Metrics, Reviews, Retrospectives, Ethics, Responsible AI Use, and AI Usage Report (Dr. Suthep)	Lab: Metrics, code review & retrospective; Responsible-AI reflection and AI Usage Report; final project hardening
14	24,26 Nov	Final Delivery, Product Demonstration, and Course Integration	Lab: Final Team Demo & Submission — product demonstration and course integration
	1–11 Dec	Final Exam (covers Weeks 8–14)	
	After Finals	Project Demonstration and Presentation	

Note: Any additional modifications to the syllabus will be announced in class.